# Enoxaparin

## Mechanism of action

- Enoxaparin is a low molecular weight heparin (LMWH) anticoagulant.
- Enoxaparin potentiates the activity of antithrombin III (a naturally occurring anticoagulant) causing inhibition of multiple coagulation factors, particularly factor Xa.

## Authorised practice levels

- Paramedics, ICPs and CCPs.

## Indications

- ✓ STEMI in conjunction with fibrinolytic therapy.

## Contraindications

- ✕ Known severe allergy.

## Cautions

- ⚠ Clinically significant bleeding. Enoxaparin will increase bleeding.
- ⚠ At risk of bleeding. If there are any cautions present within the fibrinolytic therapy checklist, personnel must seek clinical advice or discuss with the STEMI coordinator prior to administration.
- ⚠ Pregnancy.

## Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of STEMI occurring in a woman who is pregnant is so low that personnel must seek clinical advice prior to administration.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

## Dosage

- IV: 30 mg, only if the patient is aged less than 75 years.

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- SC: Dosage is based on age and known (or estimated) weight.

| Weight | Age less than 75 years | | Age 75 years or older | |
|---|---|---|---|---|
| | Enoxaparin (dose SC) | Enoxaparin (volume SC) | Enoxaparin (dose SC) | Enoxaparin (volume SC) |
| **<60 kg** | 60 mg | 0.6 ml | 45 mg | 0.45 ml |
| **60-69 kg** | 70 mg | 0.7 ml | 50 mg | 0.5 ml |
| **70-79 kg** | 80 mg | 0.8 ml | 60 mg | 0.6 ml |
| **80-89 kg** | 90 mg | 0.9 ml | 70 mg | 0.7 ml |
| **≥90 kg** | 100 mg | 1 ml | 75 mg | 0.75 ml |

## Administration

- IV:
  - Dilute 100 mg to a total of 10 ml using 0.9% sodium chloride. This gives a 10 mg/ml solution.
  - Discard 7 ml.
  - Administer 30 mg (3 ml) IV.
- SC:
  - Administer subcutaneously into the abdominal wall.
  - There is no need to sterilise the skin at the site of injection unless the skin is visibly contaminated.
  - Discard unwanted drug from the syringe before administration.
  - If an error is made in discarding unwanted drug volume and the dose remaining in the syringe is less than planned, administer the remaining dose.

## Common adverse effects

- Increased bleeding.

## Usual onset of effect

- 10-30 minutes.

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## Usual duration of effect

- 12-24 hours.

## Usual preparation

- Pre-filled syringe containing 100 mg in 1 ml.

## Pharmacokinetics

- Enoxaparin is predominately excreted in urine.
- Clearance is prolonged if the patient has significant renal impairment, but this does not alter the initial (loading) dose.

## Common interactions

- The risk of bleeding will be increased if the patient is taking an anticoagulant.

## More Information

### Version history

**3 August 2026** 1.1.0.6
**Update:** Added IV administration and dosing.

**11 September 2023** 1.0.4
**Update:** Cautions updated.